APPAJI GOUDA

Bengaluru, India | appajigouda009@gmail.com | 9448158289 | linkedin.com/in/appaji-gouda | github.com/anupgouda

PROFESSIONAL SUMMARY

AI/ML and GenAI engineer with full-stack depth. Built an end-to-end Medicare fraud risk-scoring platform (550K+ records, 0.9673 ROC-AUC, SHAP, FastAPI, PostgreSQL, 30 pytest tests), a LangChain/FAISS RAG chatbot, and a JWT/RBAC procurement system (React/TypeScript, Node.js/Express, PostgreSQL) — all deployed live. Former MERN full-stack intern. Member, TiE (The Indus Entrepreneurs).

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frontend: React, Next.js, Vite, Tailwind CSS, React Router

Backend & APIs: Node.js, Express.js, FastAPI, REST API design, JWT authentication, bcrypt, RBAC, API integration

Databases: PostgreSQL, Supabase, MongoDB, FAISS (vector search)

AI/ML: scikit-learn, Pandas/NumPy, Random Forest, Feature Engineering, Model Evaluation, SHAP (Explainable AI)

Generative AI: LangChain, RAG pipelines, Prompt Engineering, Streamlit

GenAI Coursework: Function Calling, AI Agents, Vector Databases, Fine-Tuning, SLMs, Open Models

Cloud & Tools: Vercel, Render, Streamlit Community Cloud, Git/GitHub, Postman, pytest, Google Cloud (Fundamentals)

EXPERIENCE

Full Stack Development Intern — **Nov 2024 – Jan 2025**

Mainflow Services & Technologies Pvt. Ltd. · Bengaluru, India

- Built and deployed MERN-stack (MongoDB, Express, React, Node.js) web apps used in production, from schema design to UI.
- Designed and implemented RESTful APIs, collaborating with the team on production rollouts and release cycles.
- Improved frontend responsiveness and backend performance by addressing bottlenecks found through profiling and code review.

PROJECTS

AI Chatbot with Retrieval-Augmented Generation (RAG) — **Jan 2023 – Jan 2024**

Live Demo · Source Code

- Built and deployed an end-to-end GenAI app — PDF-based semantic search chatbot (LangChain, FAISS, Streamlit), publicly accessible.
- Engineered the RAG pipeline: PDF ingestion → chunking → embeddings → FAISS vector store → semantic retrieval → LLM answer generation, grounding responses in retrieved documents to reduce unsupported generation.
- Integrated a Telegram bot to extend the chatbot to multi-channel access alongside the web UI.

College Inventory & Procurement Management System — **Aug 2023 – Oct 2023**

Live App · Backend API · Source Code

- Architected a full-stack inventory and procurement platform — React/TypeScript + Vite frontend, Node.js/Express REST API, PostgreSQL via Supabase — deployed live on Vercel and Render.
- Secured it with JWT authentication, bcrypt password hashing, and role-based access control across 6 roles (Admin, HOD, IT, Principal, Faculty, Store Manager), enforced at both frontend routes and the backend API with HTTP 403 for unauthorized requests.
- Implemented the complete procurement workflow — indents → quotations → purchase orders → bills → assets — across 10+ REST resource modules with full CRUD, status tracking, and reporting.
- Designed an isolated demo environment on a separate PostgreSQL schema so reviewers can explore the live app with sandboxed data without touching production records.

Healthcare Fraud Analytics Platform — **2025**

Live Demo · Source Code

- Engineered an end-to-end ML system scoring Medicare providers for fraud-investigation risk from 550K+ real inpatient/outpatient claims and beneficiary records: feature engineering, FastAPI prediction service, PostgreSQL persistence, Streamlit dashboard.
- Trained and tuned a Random Forest on 16 provider-level features, reaching 0.9673 ROC-AUC and 0.7864 PR-AUC on held-out data; tuned the decision threshold to 0.74 precision / 0.67 recall for an investigation-prioritization use case.
- Added SHAP explainability so every prediction ships with its feature contributions — framed as a risk signal for human investigators, not an automated fraud determination.
- Modeled an investigation workflow (status, priority, assignment, notes) on a 3-table PostgreSQL schema; validated with 30 automated tests (pytest) across the model, API, and repository layers.

EDUCATION

B.E. in Artificial Intelligence and Machine Learning — **Aug 2022 – Jul 2026**

Global Academy of Technology (VTU Affiliated) · Bengaluru, India

CERTIFICATIONS

Generative AI for Beginners — Microsoft · 21-lesson hands-on program · GitHub: github.com/anupgouda/generative-ai-for-beginners
Google Cloud Fundamentals: Core Infrastructure · Python with Pandas (Great Learning) · DevOps (Infosys) · Git Complete (Infosys) · Java Programming (TechA) · Git Foundation (TechA)